12TH BIOLOGY · CHAPTER 16 · SOLVED EXERCISE

Human Endocrine System

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Which of the following pair of hormones has antagonistic (opposing) effects?

(a) Parathyroid hormone and calcitonin (b) Growth hormone and epinephrine
(c) Estrogen and progesterone (d) Cortisol and aldosterone

Answer: (a) Parathyroid hormone and calcitonin — Parathormone raises the blood calcium level, while calcitonin lowers it.

2. Which gland produces hormones that control other endocrine glands?

(a) Anterior pituitary (b) Posterior pituitary
(c) Thyroid (d) Intermediate pituitary

Answer: (a) Anterior pituitary — Many hormones of the anterior pituitary (TSH, ACTH, LH, FSH) regulate the activity of other endocrine glands.

3. Which of the following is NOT released by anterior pituitary?

(a) Prolactin (b) ADH
(c) FSH (d) TSH

Answer: (b) ADH — ADH is made by the hypothalamus and released by the posterior pituitary.

4. Antidiuretic and oxytocin hormones are produced by

(a) Anterior pituitary (b) Posterior pituitary
(c) Intermediate pituitary (d) Hypothalamus

Answer: (d) Hypothalamus — They are made by the neurosecretory cells of the hypothalamus. The posterior pituitary only stores and releases them.

5. Which of the following hormones binds to the receptors located inside the cell?

(a) FSH (b) Testosterone
(c) Insulin (d) Growth hormone

Answer: (b) Testosterone — Testosterone is a steroid hormone. It is lipid-soluble, so it passes through the cell membrane.

6. Which of the following is a function of a hormone released by the posterior pituitary?

(a) Maturation of egg and sperm (b) Decrease in calcium levels
(c) Water retention (d) Increase in thyroid hormone level

Answer: (c) Water retention — ADH, released by the posterior pituitary, stimulates water retention by the kidneys.

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7. Which of the following hormones would be expected to increase if you were studying all day for a test and skipped breakfast and lunch?

(a) Glucagon (b) Insulin
(c) Growth hormone (d) Calcitonin

Answer: (a) Glucagon — Without food, the blood glucose level falls below normal, so glucagon is secreted to raise it.

8. Which of the following is both an endocrine and exocrine gland?

(a) Anterior pituitary (b) Pancreas
(c) Thyroid (d) Adrenal medulla

Answer: (b) Pancreas — Its exocrine cells secrete pancreatic juice into a duct, and its islets of Langerhans secrete hormones into blood.

9. Which hormone increases basal metabolic rate in the body?

(a) Thyroxine (b) Parathormone
(c) Glucagon (d) Secretin

Answer: (a) Thyroxine — Thyroid hormones stimulate cellular respiration and set the body's basal metabolic rate.

10. Which of the following is NOT a site of action of insulin?

(a) Smooth muscles (b) Skeletal muscles
(c) Adipose tissue (d) Liver

Answer: (a) Smooth muscles — Insulin acts on skeletal muscles, fat cells (adipose tissue) and the liver.

SECTION 2: SHORT QUESTIONS

1. State the role of hormones as chemical messengers.

A hormone is a regulatory chemical that is secreted into the blood or tissue fluid by an endocrine gland. The blood carries the hormone to every cell of the body, but it shows its effects only in its specific **target cells**. When the target cells receive a hormone, they respond in a specific way. That is why hormones are called **chemical messengers** or signalling molecules.

2. Where are receptors located in amino acid-based hormones and steroid hormones?

Amino acid-based hormones (peptide, protein and glycoprotein hormones, epinephrine and norepinephrine) cannot pass through the cell membrane. So their receptors are located on the **cell surface** (cell membrane) of the target cells.

Steroid hormones are lipid-soluble and can move through the cell membrane. So their receptors are located **inside the cell** (in the cytoplasm or nucleus).

According to the textbook, thyroxine and tri-iodothyronine are amino acid derivatives that can cross the cell membrane.

3. What are neurosecretory cells?

Neurosecretory cells are special cells of the hypothalamus which not only conduct nerve impulses but also secrete hormones. For example, they make oxytocin and ADH, and the releasing and inhibiting hormones that control the anterior pituitary.

4. Name the hormones of anterior pituitary gland.

1. Growth hormone (GH, or somatotropin)
2. Adrenocorticotropic hormone (ACTH, or corticotropin)
3. Thyroid-stimulating hormone (TSH, or thyrotropin)
4. Luteinizing hormone (LH)
5. Follicle-stimulating hormone (FSH)
6. Prolactin

5. Why the pituitary gland is called master gland?

The pituitary gland is called the master gland because it controls many other endocrine glands of the body. Its anterior lobe secretes hormones such as TSH, ACTH, LH and FSH, which regulate the activity of the thyroid gland, the adrenal cortex and the gonads.

6. How is the secretion of ADH controlled?

ADH is made by the neurosecretory cells of the hypothalamus. Their axons transport it to the posterior pituitary, where it is stored. So its release into the blood is controlled by the hypothalamus. When the water content of the blood is low, the hypothalamus makes the posterior pituitary release more ADH, and the kidneys retain more water. When the water content is high, less ADH is released.

The role of blood water content is taken from Chapter 14; this chapter only states that ADH is made in the hypothalamus and released by the posterior pituitary.

7. Write a note on parathyroid glands.

These are four small glands attached to the posterior surface of the thyroid gland. They secrete the hormone **parathormone**. It is released when the blood calcium level falls. Parathormone:

- stimulates the osteoclasts (bone cells) to dissolve the calcium crystals of the bone matrix and release calcium into the blood;
- stimulates the kidneys to reabsorb calcium from the urine;
- causes the activation of vitamin D, which is needed for the absorption of calcium from food in the intestine.

Over-secretion of parathormone (which may be due to a parathyroid tumour) causes excessive release of calcium from bones. It leads to deformed and soft bones. The blood calcium level rises (hypercalcemia), which depresses the nervous system and causes weakness of muscles.

Under-secretion causes hypocalcaemia. This increases the excitability of neurons. It can also lead to **tetany**, in which muscles remain in a contracted state.

8. Write the names and functions of hormones secreted by ovaries and testes.

Ovaries

1. **Estrogen:** secreted by the Graafian follicle in response to FSH. Its functions include the development of secondary sexual characteristics (e.g., breast development, widening of the pelvis, fat distribution in hips and thighs), regulation of the menstrual cycle, repair of the uterine lining after menstruation, ovulation, maintenance of bone density, lowering of "bad" cholesterol (LDL), maintenance of healthy blood pressure, and regulation of mood and memory.
2. **Progesterone:** produced by the corpus luteum in response to LH during the menstrual cycle, and by the placenta during pregnancy. It inhibits further FSH secretion from the pituitary, so no more follicles ripen. It causes further thickening and vascularisation of the uterus wall for maintaining pregnancy. It suppresses ovulation.

Testes

LH stimulates the testes to secrete a group of sex hormones called androgens. **Testosterone** is an androgen secreted by the interstitial cells among the seminiferous tubules. It initiates the maturation of male reproductive organs and the appearance of secondary sex characteristics and sex drive. It is also necessary for normal sperm production.

9. Differentiate between;

• Exocrine and endocrine glands

Exocrine glands	Endocrine glands
They have ducts.	They are ductless.
They release their secretions into the ducts attached with them.	They release their secretions (hormones) directly into blood or tissue fluid.
Example: exocrine cells of pancreas (pancreatic juice).	Examples: pituitary, thyroid, adrenal glands.

• Hypothyroidism and hyperthyroidism

Hypothyroidism	Hyperthyroidism
It is the deficiency of thyroid hormones.	It is the excessive secretion of thyroid hormones.
It causes weight gain and lethargy.	It causes weight loss.
Heart rate and body temperature are low.	Blood pressure, heart rate and body temperature are high.
It may result in cretinism, myxoedema or goitre.	It may be caused by Graves' disease.

• Parathormone and calcitonin

Parathormone	Calcitonin
It is secreted by the parathyroid glands.	It is secreted by the thyroid gland.
It is released when the blood calcium level falls.	It is released when the blood calcium level rises too high.
It releases calcium from bones into the blood and raises the blood calcium level.	It stimulates the absorption of calcium into bones and lowers the blood calcium level.

• Beta and alpha cells of pancreas

Beta (β) cells	Alpha (α) cells
They are larger in number in each islet.	They are fewer in number.
They secrete insulin.	They secrete glucagon.
They release their hormone when blood glucose goes above normal.	They release their hormone when blood glucose goes below normal.

• Insulin and glucagon

Insulin	Glucagon
It is secreted by the beta cells of the islets of Langerhans.	It is secreted by the alpha cells of the islets of Langerhans.
It is secreted when the blood glucose level rises.	It is secreted when the blood glucose level is below normal.
It reduces the blood glucose level.	It raises the blood glucose level.
It promotes the conversion of glucose to glycogen in the liver.	It promotes the hydrolysis of stored glycogen in the liver.

• Diabetes insipidus and diabetes mellitus

Diabetes insipidus	Diabetes mellitus
It is caused by the under-secretion of ADH.	It is caused by the under-secretion of insulin.
The blood glucose level is not affected.	The blood glucose level is above normal (hyperglycaemia).
Excessive dilute urine is produced, with frequent thirst.	Large amounts of urine are produced, leading to dehydration and kidney damage.

• Estrogen and progesterone

Estrogen	Progesterone
It is secreted by the Graafian follicle.	It is produced by the corpus luteum (and by the placenta during pregnancy).
It is secreted in response to FSH.	It is produced in response to LH.
It develops secondary sexual characteristics and repairs the uterine lining after menstruation.	It causes further thickening and vascularisation of the uterus wall for maintaining pregnancy.
It has a role in ovulation.	It suppresses ovulation.

• Positive and negative feedback

Positive feedback	Negative feedback
The product or response speeds up the action of the controlling system.	The end product stops the initial step of the process.
It is less common.	It is common in the endocrine system.
Example: release of oxytocin during childbirth.	Example: control of insulin and glucagon secretion.

SECTION 3: LONG QUESTIONS

1. Explain the neurosecretory role of hypothalamus.

Many endocrine glands of the body are controlled by the master gland, i.e., the pituitary gland. However, the master gland is itself controlled by a part of the brain, i.e., the **hypothalamus**. The hypothalamus contains special **neurosecretory cells**, which not only conduct nerve impulses but also secrete hormones.

Control of posterior pituitary

Two hormones, oxytocin and antidiuretic hormone (ADH), are made by the neurosecretory cells of the hypothalamus. The axons of these cells transport oxytocin and ADH to the posterior pituitary, where they are stored for eventual release into the blood.

Control of anterior pituitary

Blood vessels connect the hypothalamus with the anterior pituitary. Neurosecretory cells of the hypothalamus secrete hormones that travel to the anterior pituitary through these blood vessels. These hormones are of two kinds:

- **Releasing hormones** stimulate the anterior pituitary to make and secrete hormones.
- **Inhibiting hormones** inhibit the production and secretion of anterior pituitary hormones.

Hormone of hypothalamus	Effect on anterior pituitary
Growth hormone releasing hormone	Secretion of growth hormone
Somatostatin	Inhibition of growth hormone
Thyrotropin releasing hormone (TRH)	Secretion of thyroid stimulating hormone (TSH)
Adreno-corticotropin releasing hormone	Secretion of adrenocorticotrophic hormone
Prolactin inhibiting factor (actually dopamine)	Inhibits secretion of prolactin
Gonadotropin releasing hormone	Secretion of FSH and LH

2. Describe the chemical nature of hormones.

Hormones are grouped into the following chemical categories.

1. Peptide and protein hormones

These are made of chains of amino acids. They are water-soluble, so they dissolve easily in blood. However, they cannot pass through cell membranes. So they bind to receptors on the cell surface.

Hormone	Source (gland)	Main function
Insulin	Pancreas	Lowers blood glucose level
Glucagon	Pancreas	Raises blood glucose level
Growth hormone	Anterior pituitary	Stimulates growth and cell reproduction
Oxytocin	Hypothalamus (released by posterior pituitary)	Stimulates muscle contractions during childbirth

2. Glycoprotein hormones

Glycoproteins are conjugated proteins. They consist of a protein combined with a carbohydrate (sugar) group. They are large, complex and polar molecules, so they also cannot pass through the cell membrane.

Hormone	Source (gland)	Main function
Thyroid stimulating hormone	Anterior pituitary	Stimulates the thyroid gland to produce thyroxine
Follicle stimulating hormone	Anterior pituitary	Stimulates sperm production or egg development
Luteinizing hormone	Anterior pituitary	Triggers ovulation in females and secretion of testosterone in males

3. Amino acid derivative hormones

These are small molecules derived from a single amino acid (usually tyrosine or tryptophan). Their ability to pass through cell membranes depends on whether they behave more like a protein or more like a lipid. Epinephrine and norepinephrine cannot cross membranes, while thyroxine and tri-iodothyronine can cross.

Hormone	Source (gland)	Main function
Epinephrine (adrenaline)	Adrenal medulla	Immediate “fight or flight” response
Norepinephrine (noradrenaline)	Adrenal medulla	Increases blood pressure and alertness
Thyroxine (T_4)	Thyroid gland	Controls the speed of metabolism
Triiodothyronine (T_3)	Thyroid gland	Regulates body temperature and heart rate

4. Steroid hormones

These hormones are derived from cholesterol (a type of lipid). Unlike proteins, steroids are lipid-soluble (fat-soluble). So they can move through the cell membrane and give instructions directly to the nucleus.

Hormone	Source (gland)	Main function
Cortisol	Adrenal cortex	Regulates metabolism and helps the body respond to stress
Aldosterone	Adrenal cortex	Helps the kidneys manage salt and water balance
Testosterone	Testes	Regulates male reproductive tissues and traits
Estrogen / Progesterone	Ovaries	Regulate the female reproductive system and cycle

3. State the names of the hormones secreted by pituitary gland. Write the function of each hormone and state the major abnormalities caused by the imbalance of pituitary hormones.

The pituitary gland is attached to the hypothalamus by a stalk called the infundibulum. It has three lobes.

Hormones of anterior pituitary

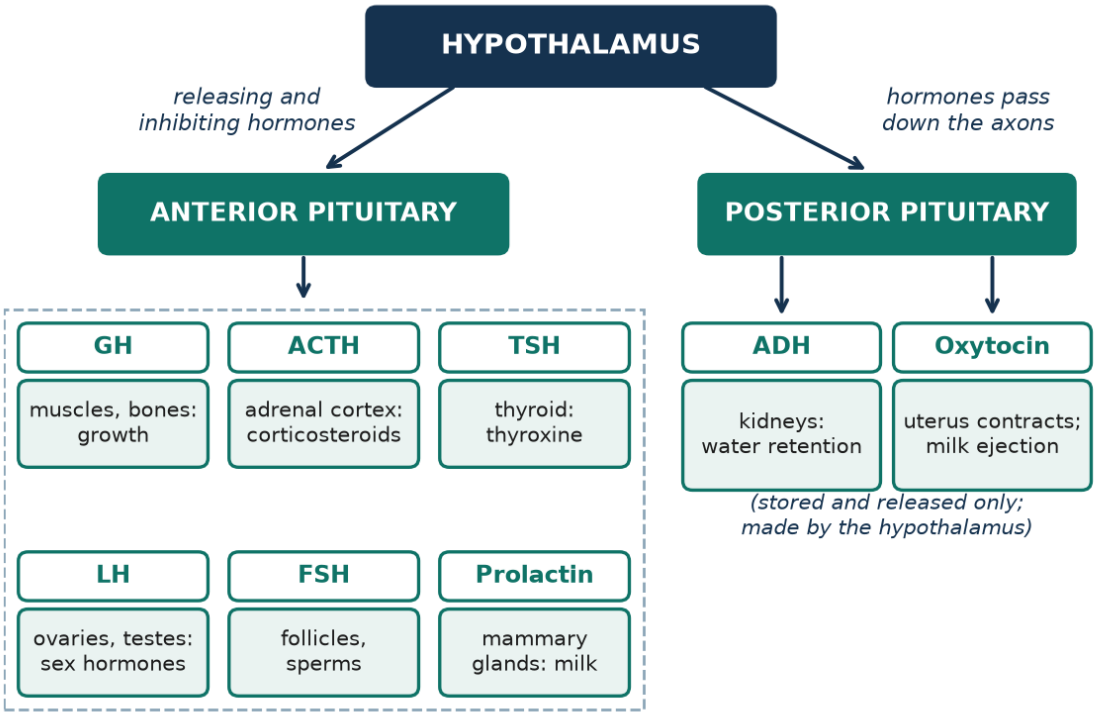
1. **Growth hormone (GH, or somatotropin):** stimulates the growth of muscles, bones (indirectly) and other tissues. It is also essential for proper metabolic regulation.
2. **Adrenocorticotropic hormone (ACTH, or corticotropin):** stimulates the adrenal cortex to secrete corticosteroids (cortisone and aldosterone), which regulate glucose and salt homeostasis.

3. **Thyroid-stimulating hormone (TSH, or thyrotropin):** stimulates the thyroid gland to produce thyroxine, which in turn stimulates oxidative respiration.
4. **Luteinizing hormone (LH):** in females, it stimulates the ovaries to produce progesterone and estrogen and also initiates ovulation. In males, it stimulates the testes to produce testosterone.
5. **Follicle-stimulating hormone (FSH):** stimulates the development of ovarian follicles in females and the development of sperms in males. FSH and LH are both called gonadotropins.
6. **Prolactin:** stimulates the mammary glands to produce milk. It is produced in both males and females but functions only in females.

Hormones of posterior pituitary

The posterior pituitary contains axons that originate in the hypothalamus. It does not synthesize any hormone; it only stores and releases the hormones made by the hypothalamus:

1. **Antidiuretic hormone (ADH or vasopressin):** stimulates water retention by the kidneys.
2. **Oxytocin:** stimulates milk ejection by contracting the smooth muscles of the mammary glands. It also stimulates uterine contractions in women during childbirth.



Hormones of the pituitary gland and their targets

Hormone of intermediate pituitary

In humans, the intermediate pituitary is a thin layer of cells between the anterior and posterior pituitary. It produces **melanocyte stimulating hormone (MSH)**, which stimulates the production and release of melanin by melanocytes in skin and hair.

Imbalance of pituitary hormones

1. **Gigantism:** over-secretion of GH in a growing child. The height is well above average, with enlarged hands, feet and facial features.
2. **Acromegaly:** over-secretion of GH in adult life. Bones can no longer increase in length but grow in thickness. Hands, feet, skull, nose and jawbone become enlarged.
3. **Dwarfism:** deficiency of GH. Development is much slower and the individual has short stature. However, the body parts stay in proportion, and brain development and IQ are unaffected.
4. **TSH:** over-secretion causes hyperthyroidism (excess of thyroxine) and under-secretion causes hypothyroidism (lack of thyroxine).
5. **ADH:** under-secretion causes diabetes insipidus (excessive production of dilute urine and frequent thirst). Over-secretion may lead to kidney problems.
6. **Oxytocin:** over-secretion causes rupturing of the uterine wall, while under-secretion inhibits the normal labour process.

4. Describe the functions of thyroid hormones. What are major diseases due to their imbalance?

The thyroid is a butterfly-shaped gland located on either side of the trachea in the front of the neck. It is composed of two connected lobes. It secretes the following hormones:

1. **Tetra-iodothyronine (T_4) or thyroxine and tri-iodothyronine (T_3):** The thyroid produces much more T_4, which is eventually converted into T_3, the more active and powerful form. These hormones increase heart rate and breathing rate. They stimulate cellular respiration and set the body's basal metabolic rate. They also enhance glucose catabolism and the synthesis of cholesterol in the liver. In children, they promote the growth of the skeleton and the development of the brain.
2. **Calcitonin:** It maintains the proper level of calcium in blood. When the blood calcium concentration rises too high, calcitonin stimulates the absorption of calcium into bones, thus lowering its level in blood.

Imbalance of thyroid hormones

1. **Hypothyroidism** (deficiency of thyroid hormones): causes growth retardation, lethargy, weight gain, constipation, high cholesterol, and low heart rate and body temperature.
2. **Cretinism:** hypothyroidism during foetal and childhood development; a condition of stunted growth and mental retardation.
3. **Myxoedema:** a rare, life-threatening complication of severe hypothyroidism. Mucopolysaccharides build up in the connective tissues, causing swelling and thickening of the skin and tissues.
4. **Goitre:** swelling of the thyroid gland; it results when hypothyroidism is caused by iodine deficiency.
5. **Hyperthyroidism** (excessive secretion of thyroid hormones): causes weight loss, and high blood pressure, heart rate and body temperature.

5. Write a detailed note on adrenal glands mentioning the hormones of adrenal context and adrenal medulla and their functions.

"Adrenal context" is printed in the textbook; it means adrenal cortex.

One adrenal gland is located just above each kidney. Each gland is composed of an inner portion, the **adrenal medulla**, and an outer layer, the **adrenal cortex**.

Adrenal medulla

It produces two hormones: **epinephrine** (adrenaline) and **norepinephrine** (noradrenaline). When a person is stressed, the medulla secretes these hormones. They prepare the body for "fight or flight". In this response there is increased heart rate, increased blood pressure, dilation of the bronchioles, elevation in blood glucose, and reduced blood flow to the skin and digestive organs.

Adrenal cortex

The hormones of the adrenal cortex are called **corticosteroids**. These include:

1. **Glucocorticoids**, e.g., cortisol and related steroids. They maintain glucose homeostasis. They stimulate the breakdown of muscle protein into amino acids, which are carried by blood to the liver. They also stimulate the liver for gluconeogenesis (conversion of amino acids into glucose).
2. **Mineralocorticoids**, e.g., aldosterone. It helps to regulate mineral balance in blood. It stimulates the kidneys to reabsorb sodium from urine; in this way water is also reabsorbed, so normal blood volume and pressure are maintained. It also stimulates the kidneys to secrete the excess of potassium into urine.

Imbalance of adrenal gland hormones

1. **Adrenal medulla:** over-secretion may cause hypertension and aggressive behaviour during routine life. Under-secretion causes failure to combat an emergency situation.
2. **Cushing disease:** over-secretion of cortisol. It occurs when the pituitary gland produces too much ACTH. There is excessive protein breakdown. It results in high blood sugar, high blood pressure, thin bones, muscle loss, thin and fragile skin, depression and too much facial hair (in women).
3. **Addison's disease:** under-secretion of cortisol. It is characterized by general metabolic disturbance, weakness of muscles and loss of salts. The symptoms usually appear slowly.

6. Write a note on pancreas as an endocrine gland.

The pancreas mostly contains exocrine cells, which secrete pancreatic juice into the pancreatic duct. But it also has specialized cells which function as an endocrine gland. These cells are distributed as patches called **islets of Langerhans**. Each islet has two types of cells: **beta (β) cells**, which are larger in number and secrete insulin, and **alpha (α) cells**, which secrete glucagon.

Insulin reduces the blood glucose level. It is secreted when the level of blood glucose rises. It increases the rate of glucose uptake by body cells (e.g., skeletal muscles and fat cells). It promotes glycogenesis (conversion of glucose to glycogen) in the liver,

increases the use of glucose in cellular respiration, promotes the conversion of excess glucose to fats and inhibits gluconeogenesis (glucose synthesis).

Glucagon raises the blood glucose level. It is secreted when the blood glucose level is below normal. It mainly promotes the hydrolysis of stored glycogen in the liver. As a result, glucose is released into the blood and can be taken up by cells and used for energy.

Imbalance of pancreas hormones

1. **Diabetes mellitus (hyperglycaemia):** caused by under-secretion of insulin. Cells are unable to obtain glucose from blood, so the blood glucose level is above normal. The excess glucose inhibits water reabsorption by the kidneys, so large amounts of urine are produced. It leads to dehydration and kidney damage.
2. **Hypoglycaemia:** caused by over-secretion of insulin. The blood glucose level falls below normal. Symptoms include lethargy, dizziness, nervousness and, in extreme cases, unconsciousness.

7. Describe the functions of the hormones secreted by thymus glans, pineal gland, kidneys, and walls of digestive tract.

"Thymus glans" is printed in the textbook; it means thymus gland.

Gland / tissue	Hormone	Function
Thymus gland (beneath the sternum, between the lungs)	Thymosin	Stimulates the maturation of T cells, which help defend the body from pathogens
Pineal gland (near the base of brain)	Melatonin	Regulates sleep patterns
Stomach wall	Gastrin	Stimulates the gastric glands to release gastric juice
Small intestine wall	Secretin	Stimulates the pancreas to release pancreatic juice
Small intestine wall	Cholecystokinin	Stimulates the gallbladder to release bile
Kidneys	Renin	Controls the production of aldosterone in adrenal glands
Kidneys	Erythropoietin	Stimulates bone marrow for the production of RBCs
Kidneys	Calcitriol	Increases blood calcium level by absorption of calcium from intestines

INQUISITIVE QUESTIONS

1. State the role of artificially synthesized steroids in sports and their long-term effects on its users.

Artificially synthesized steroids, also known as **anabolic steroids**, are synthetic versions of testosterone. Athletes have widely used them to enhance performance in sports, because they increase muscle mass, strength and endurance. However, their use in sports is illegal and banned. Their long-term effects are:

- Damage to the liver, leading to hepatitis and liver tumours.
- Increased risk of heart disease, stroke and blood clots.
- Disruption of the body's natural hormone balance, leading to infertility, impotence and breast development in men.
- Mental health issues such as depression, anxiety and aggressive behaviour.

2. Explain on what grounds some companies claim that growth is possible in people having short heights.

These companies base their claim on the action of **growth hormone (GH)**. GH stimulates the growth of muscles, bones and other tissues, and its deficiency causes dwarfism. GH treatment is given to children who have short stature due to GH deficiency, and it improves their growth rate. So the companies claim that their products can increase height.

However, this claim is not true for everyone. Every short person does not have GH deficiency, so GH therapy may not be appropriate for everyone. Also, in adults the bones are no longer capable of increasing in length; extra GH only makes them thicker (acromegaly).

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